

A counterexample, exact spacing criteria, and a sharp smoothing threshold for products of q -integers

Prepared with ChatGPT; self-contained proofs and exact computational checks

Abstract

We attack Conjecture 5.4 of Connelly, Ito, Martinez, Shevchenko, and Yang, *Unimodality of q -Fibonomial coefficients for small cases*, arXiv:2605.12822v1. Its necessity assertion is false: $(1 + q)^6(1 + q^3)$ is unimodal but violates the proposed condition. We then prove the conjectured sufficient condition for every spacing $r \geq 2$. For spacing three we obtain a complete replacement: if no a_i is divisible by three, then

$$\left(\prod_i [a_i]_q \right) [b]_{q^3} \text{ is unimodal} \iff b \leq 1 + \sum_i \left\lfloor \frac{a_i}{3} \right\rfloor + 2 \left\lfloor \frac{\#\{i : a_i \equiv 2 \pmod{3}\}}{6} \right\rfloor.$$

A failed inequality produces a specified first-half coefficient drop of exactly one. We also establish a general necessary degree bound and settle the spacing-two criterion. A dedicated study of the two-shift family $(1 + q)^n(1 + q^r)$ proves the sharp threshold “unimodal if and only if $n \geq r^2 - 3$ ” in two independent ways, determines the equality case (exactly four consecutive maximal coefficients at the threshold), shows that the spacing-three counterexamples are log-concave for every $n \geq 6$, and records a strict-maximum witness $(1 + q)^9(1 + q^3)$. A complete finite search over *all* spacings proves that degree nine is the minimum possible degree of a counterexample and that its parameter set is unique. The arguments are finite and algebraic; computations are supplementary. The main q -Fibonomial unimodality conjecture is not resolved here.

The counterexample at a glance, and a strict companion

$$(1 + q)^6(1 + q^3) = 1 + 6q + 15q^2 + 21q^3 + 21q^4 + 21q^5 + 21q^6 + 15q^7 + 6q^8 + q^9.$$

Here $r = 3$, $b = 2$, and $a_1 = \dots = a_6 = 2$. No a_i is divisible by 3, and $1 + \sum_i \lfloor a_i/3 \rfloor = 1 < 2$. Nevertheless, the displayed coefficient sequence is weakly unimodal. It is in fact log-concave, and three further factors $1 + q$ give

$$(1 + q)^9(1 + q^3) = 1 + 9q + 36q^2 + 85q^3 + 135q^4 + 162q^5 + 168q^6 + 162q^7 + 135q^8 + 85q^9 + 36q^{10} + 9q^{11} + q^{12},$$

which has a *unique* maximum and strictly monotone flanks while still violating the proposed condition. The refutation is therefore not an artefact of the central plateau. Both witnesses are carried throughout (item D5 of Table 6).

Status. The statements below are accompanied by proofs and reproducible integer-arithmetic tests. They have not been independently refereed or checked by a proof assistant. The counterexample refutes the specified source statement; historical priority of the counterexample and the stronger theorems has not been established by an exhaustive literature review.

1 The selected conjecture and the result

For a positive integer a , put

$$[a]_q = 1 + q + \dots + q^{a-1}.$$

Throughout, q is a formal indeterminate, $r \geq 2$, $k \geq 1$, and $a_1, \dots, a_k, b$ are positive integers. We study

$$P_{a,b,r}(q) = \left(\prod_{i=1}^k [a_i]_q \right) [b]_{q^r}. \quad (1.1)$$

Its coefficients are nonnegative integers, its constant and leading coefficients are one, and its degree is

$$D = S + r(b - 1), \quad S = \sum_{i=1}^k (a_i - 1). \quad (1.2)$$

Every factor is symmetric, so P is symmetric.

The source of the attack is [1, Section 5.2, Conjecture 5.4, p. 14]. For brevity write its proposed condition as

$$C_r(a, b): \quad (\exists i: r \mid a_i) \quad \text{or} \quad b \leq 1 + \sum_i \left\lfloor \frac{a_i}{r} \right\rfloor. \quad (1.3)$$

The conjecture asserts that (1.3) is sufficient, and also necessary when $k \leq 3$ or $r \leq 3$. The distinction between “or” and “and” matters: the case $k = 6, r = 3$ lies inside the asserted necessity range. The source reports tests with $k \leq 5, r \leq 6$, and $\max\{a_i, b\} \leq 15$ [1, p. 14]. This is an auxiliary conjecture inside a paper about q -Fibonomial coefficients; it is not the original Fibonomial unimodality conjecture of [2].

Proposition 1.1 (Exact disproof). *The necessity assertion in [1, Conjecture 5.4] is false.*

Proof. Take $r = 3, b = 2$, and $a_i = 2$ for $1 \leq i \leq 6$. The binomial theorem gives the coefficient list

$$(1, 6, 15, 21, 21, 21, 15, 6, 1).$$

It increases weakly to its central plateau and then decreases weakly. On the other hand, $3 \nmid 2$ and $b = 2 > 1 + 6\lfloor 2/3 \rfloor = 1$, so (1.3) fails. Since $r = 3$, the claimed necessity applies to this example. $\square$

The complete local certificate is small enough to print in full:

j	0	1	2	3	4	5	6	7	8	9
$[q^j]P$	1	6	15	21	21	21	21	15	6	1
Difference from previous	—	5	9	6	0	0	0	-6	-9	-5

This is a counterexample to an auxiliary product criterion, not to the main q -Fibonomial conjecture in [1]. Our stronger conclusions, proved below, are summarized in Table 1. All appearances of “unimodal” allow equal consecutive coefficients; a sequence is unimodal if it is weakly increasing up to some index and weakly decreasing thereafter, intervening zeros included.

How this note is organized. Sections 2–6 treat the general product (1.1): closure tools, the centered peeling identity and sufficiency, a residue-mass obstruction and a necessary degree bound, and complete criteria for spacings two and three. Section 7 is a deep dive on the two-shift slice $b = 2, a_i = 2$, which is exactly the family containing the counterexample. Section 8 settles minimum degree across all spacings, Section 9 gives a clearly labelled probabilistic heuristic for the quadratic scale, Section 10 records the computations, Section 11 the provenance, and Section 12 what remains open.

Setting	Conclusion proved in this note
Every $r \geq 2$	The condition C_r is sufficient (Theorem 3.2). With no divisible factor, unimodality also requires $b \leq 1 + \lfloor S/r \rfloor$ (Theorem 4.2).
$r = 2$	The original condition C_2 is necessary and sufficient (Theorem 5.1).
$r = 3$	With no divisible factor, the exact upper bound on b is $1 + \sum_i \lfloor a_i/3 \rfloor + 2\lfloor t/6 \rfloor$, where $t = \#\{i : a_i \equiv 2 \pmod{3}\}$ (Theorem 6.1).
Two shifted binomial rows	$(1+q)^n(1+q^r)$ is unimodal exactly when $n \geq r^2 - 3$, for $r \geq 2$, $n \geq 0$ (Theorem 7.1, with two independent proofs). At the threshold exactly four consecutive coefficients are maximal (Corollary 7.6).
Gap three, all $n \geq 6$	$(1+q)^n(1+q^3)$ is a <i>log-concave</i> counterexample (Corollary 7.8); and for every $B \geq 2$, $(1+q)^t[B]_{q^3}$ with $t \geq 6\lfloor B/2 \rfloor$ is one as well (Corollary 6.3).
All spacings	Minimum counterexample degree is nine, with a unique normalized parameter set (Proposition 8.1).

Table 1: Mathematical results, rather than extrapolations from the finite tests.

2 Elementary tools for symmetric unimodality

A finite nonnegative sequence $(c_0, \dots, c_D)$ is *unimodal* if, for some index m , it is weakly increasing up to m and weakly decreasing after m . It is *symmetric* if $c_j = c_{D-j}$. For a symmetric sequence, unimodality is equivalent to

$$c_j - c_{j-1} \geq 0 \quad 0 \leq j \leq \lfloor D/2 \rfloor, \quad c_{-1} = 0. \quad (2.1)$$

Indeed, these inequalities control the left half and reflection controls the right half; in odd degree the two central entries agree automatically. We abbreviate “symmetric and unimodal” by SU. All coefficient sequences are extended by zero outside their support.

A shifted sequence $q^s H(q)$ has its center shifted by s . In a sum of shifted polynomials, symmetry is understood about the *specified common center*, not automatically half the largest exponent of each summand. This convention is essential in the decomposition used below.

Lemma 2.1 (Centered intervals and closure). *Products of SU polynomials with nonnegative coefficients are SU. Nonnegative sums of SU coefficient sequences with the same center are also SU.*

Proof. If $F(q) = \sum_{j=0}^D f_j q^j$ is SU, let $\alpha_j = f_j - f_{j-1}$ for $0 \leq j \leq \lfloor D/2 \rfloor$, with $f_{-1} = 0$. Then $\alpha_j \geq 0$ and

$$F(q) = \sum_{j=0}^{\lfloor D/2 \rfloor} \alpha_j q^j [D - 2j + 1]_q. \quad (2.2)$$

To verify the identity, the coefficient at q^m on the right is $\sum_{j=0}^{\min(m, D-m)} \alpha_j = f_{\min(m, D-m)} = f_m$. Each summand is the indicator sequence of an interval centered at $D/2$.

For positive u, v , the coefficients of $[u]_q [v]_q$ at exponents $0 \leq \ell \leq u + v - 2$ are

$$\min\{\ell + 1, u, v, u + v - 1 - \ell\}.$$

They form a symmetric triangular or trapezoidal sequence. Apply (2.2) to two factors of degrees D and E . Every product of two interval summands is SU with center $(D + E)/2$. On the left of this center, each first difference is nonnegative, so the same is true of their sum. Symmetry supplies the right half. This also proves the assertion about sums with a common center. $\square$

The special case in which one factor is $1 + q$ admits a two-line proof that avoids the interval decomposition entirely. We record it separately because it is the only closure fact the two-shift analysis of Section 7 needs, and because it also yields the plateau bookkeeping of Section 7.6.

Lemma 2.2 ((1 + q)-smoothing). *If $A(q)$ has nonnegative, symmetric, unimodal coefficients, then so does $(1 + q)A(q)$.*

Proof. Write $A(q) = \sum_{j=0}^D a_j q^j$, extending a_j by zero outside its support. The new coefficients satisfy

$$b_j = a_j + a_{j-1}, \quad b_j - b_{j-1} = a_j - a_{j-2}. \quad (2.3)$$

Symmetry and nonnegativity are immediate. For j in the left half of the new sequence, $a_j \geq a_{j-2}$ follows from the old left-half monotonicity, except possibly at the new middle index when $D = 2h + 1$. At that index $j = h + 1$, and symmetry gives $a_{h+1} = a_h \geq a_{h-1}$, which is the required inequality. Now apply (2.1). $\square$

Lemmas 2.1 and 2.2 are kept as a deliberate pair; see item D10 of Table 6.

Lemma 2.3 (A divisible factor). *If $r \mid a_i$ for some i , then $P_{a,b,r}$ is unimodal for every positive integer b .*

Proof. Write $a_i = rA$. The identity

$$[rA]_q = [r]_q [A]_{q^r}$$

shows that the relevant two factors are

$$[r]_q [A]_{q^r} [b]_{q^r}.$$

The polynomial $[A]_z [b]_z$ has an SU coefficient sequence in z . Replacing z by q^r and multiplying by $[r]_q$ repeats each coefficient exactly r times. Repetition preserves symmetry and unimodality. The remaining factors are ordinary q -integers, so Lemma 2.1 finishes the proof. $\square$

3 A centered identity proves the sufficient condition

The following identity removes one block of length r from an ordinary factor and one term from the spaced factor, without changing the center of the residual sequence.

Lemma 3.1 (Centered peeling). *For integers $a > r$ and $b \geq 2$,*

$$[a]_q [b]_{q^r} = [a + r(b - 1)]_q + q^r [a - r]_q [b - 1]_{q^r}. \quad (3.1)$$

Both terms on the right, with zero padding, are symmetric about the center of the polynomial on the left.

Proof. Multiply (3.1) by $(1 - q)(1 - q^r)$. The identity becomes

$$\begin{aligned} (1 - q^a)(1 - q^{rb}) &= (1 - q^{a+r(b-1)})(1 - q^r) \\ &\quad + q^r (1 - q^{a-r})(1 - q^{r(b-1)}), \end{aligned}$$

which follows on expansion. The left side of (3.1) has polynomial degree

$$d = a - 1 + r(b - 1).$$

The first term on the right has degree d . The unshifted residual polynomial has degree $d - 2r$, so its shift by q^r has center $r + (d - 2r)/2 = d/2$. $\square$

Theorem 3.2 (Sufficiency for every spacing). *For every $r \geq 2$ and every number k of ordinary factors, $C_r(a, b)$ implies unimodality of $P_{a,b,r}$.*

Proof. A divisible factor is covered by Lemma 2.3. Assume there is none, and write

$$Q = \sum_i \left\lfloor \frac{a_i}{r} \right\rfloor.$$

We prove sufficiency of $b \leq Q + 1$ by induction on b . For $b = 1$, the spaced factor equals one and the product of ordinary q -integers is SU by Lemma 2.1.

Let $b \geq 2$. Then $Q \geq 1$, so at least one a_i is greater than r : equality is impossible because there is no divisible factor. Apply (3.1) to this factor and multiply by all the others. The first summand is a product of ordinary q -integers, hence SU. In the residual product, a_i is replaced by $a_i - r$ and b by $b - 1$. No new divisible factor appears, the floor sum becomes $Q - 1$, and

$$b - 1 \leq Q = (Q - 1) + 1.$$

The induction hypothesis makes the residual product SU. Lemma 3.1 shows that the shifted residual and the first summand have the same center, even after multiplication by the other factors. Their sum is therefore SU by Lemma 2.1. $\square$

This settles the half of [1, Conjecture 5.4] that the counterexample does not touch. It is proved here, not merely tested; the degree-nine exhaustive search of Section 8 independently found no violation of sufficiency, which is a corroborating bounded observation rather than evidence for the theorem.

Corollary 3.3 (Reduction to small residues). *Suppose no a_i is divisible by r , and write*

$$a_i = rA_i + s_i, \quad 1 \leq s_i \leq r - 1, \quad Q = \sum_i A_i.$$

If $b > Q + 1$ and

$$\left(\prod_i [s_i]_q \right) [b - Q]_{q^r} \tag{3.2}$$

is SU, then $P_{a,b,r}$ is SU.

Proof. Apply (3.1) a total of Q times, subtracting r from some a_i at each step. The spaced parameter never drops below two, since its final value is $b - Q \geq 2$. The final residual is (3.2). Reversing the steps, each step adds an SU product of ordinary q -integers at the same center as the shifted residual. $\square$

Corollary 3.3 is a sufficient reduction, not an assertion that residue reduction preserves unimodality in both directions for every r . That converse will be obtained for $r = 3$ by a separate obstruction argument.

4 Periodic coefficient differences and a necessary bound

Let

$$F(q) = \prod_i [a_i]_q = \sum_{j=0}^S f_j q^j, \quad R_s = \sum_{\substack{0 \leq j \leq S \\ j \equiv s \pmod{r}}} f_j, \quad 0 \leq s < r.$$

The R_s are the total coefficient masses in the residue classes modulo r . Put $\delta_s = R_s - R_{s-1}$, with subscripts interpreted modulo r . As a formal power series, define

$$U(q) = \frac{F(q)}{1 - q^r} = \sum_{n \geq 0} u_n q^n, \quad u_n = \sum_{\ell \geq 0} f_{n-r\ell}. \tag{4.1}$$

As usual, coefficients outside their support are zero.

Lemma 4.1 (The periodic tail). *If $n \geq \max\{1, S - r + 2\}$, then $u_n - u_{n-1} = \delta_{n \bmod r}$. If also $n < rb$, the same difference occurs in $P = F[b]_{q^r}$:*

$$[q^n]P - [q^{n-1}]P = \delta_{n \bmod r}. \quad (4.2)$$

Proof. For the stated n , the next possible omitted indices in the residue classes of n and $n - 1$ are $n + r > S$ and $n - 1 + r > S$. Consequently $u_n = R_{n \bmod r}$ and $u_{n-1} = R_{(n-1) \bmod r}$. Finally,

$$P(q) = (1 - q^{rb})U(q),$$

so its coefficients agree with those of U below exponent rb . $\square$

Theorem 4.2 (A necessary degree bound for every spacing). *Suppose no a_i is divisible by r . If $P_{a,b,r}$ is unimodal, then*

$$r(b-1) \leq S, \quad \text{equivalently} \quad b \leq 1 + \left\lfloor \frac{S}{r} \right\rfloor. \quad (4.3)$$

Proof. Let ζ be a primitive r th root of unity. Every factor

$$[a_i]_q \Big|_{q=\zeta} = \frac{1 - \zeta^{a_i}}{1 - \zeta}$$

is nonzero, so $F(\zeta) \neq 0$. The R_s cannot all be equal, since otherwise $F(\zeta) = R_0(1 + \zeta + \dots + \zeta^{r-1}) = 0$. The differences δ_s sum to zero and are not all zero, so at least one is negative.

Choose an integer n in the r -term interval

$$S - r + 2 \leq n \leq S + 1$$

whose residue has a negative difference. Such an n is automatically positive. Indeed, if the interval includes nonpositive integers then $S \leq r - 2$; at $n = 0$ the difference is $R_0 - R_{r-1} = f_0 > 0$, and for any negative n in this interval both relevant residue masses vanish. Thus Lemma 4.1 applies.

Suppose for a contradiction that $L = r(b-1) > S$. Then $n \leq S + 1 < rb$. If $L \geq S + 2$, we also have $n \leq S + 1 \leq \lfloor (S + L)/2 \rfloor$. Equation (4.2) gives a negative first-half difference, contradicting unimodality.

The remaining possibility is $L = S + 1$. In that case $S \equiv -1 \pmod{r}$. Symmetry of F implies $R_0 = R_{r-1}$, hence $\delta_0 = 0$. The selected negative difference cannot be at $n = S + 1$, whose residue is zero. Therefore $n \leq S = \lfloor (S + L)/2 \rfloor$, and again (4.2) contradicts unimodality. $\square$

Together, Theorems 3.2 and 4.2 give a useful sandwich. When there is no divisible factor,

$$b \leq 1 + \sum_i \left\lfloor \frac{a_i}{r} \right\rfloor \implies P \text{ unimodal} \implies b \leq 1 + \left\lfloor \frac{\sum_i (a_i - 1)}{r} \right\rfloor. \quad (4.4)$$

The residue classes decide what happens in the possible gap.

5 Spacing two: the original criterion is exact

Theorem 5.1 (Complete spacing-two criterion). *The polynomial $(\prod_i [a_i]_q)[b]_{q^2}$ is unimodal if and only if some a_i is even or*

$$b \leq 1 + \sum_i \left\lfloor \frac{a_i}{2} \right\rfloor.$$

Proof. Sufficiency follows from Theorem 3.2. If all a_i are odd, write $a_i = 2A_i + 1$ and $Q = \sum_i A_i$. Then $S = 2Q$, so Theorem 4.2 gives the same necessary bound $b \leq Q + 1$. $\square$

There is also an exact failure certificate. With all a_i odd, $F(-1) = 1$, so the even residue mass exceeds the odd residue mass by one. For $b \geq Q + 2$, the index $n = S + 1 = 2Q + 1$ lies below $2b$ and no later than the center $Q + b - 1$. Lemma 4.1 gives

$$[q^{2Q+1}]P - [q^{2Q}]P = -1.$$

This argument also covers factors $a_i = 1$.

6 Spacing three: a complete replacement theorem

Assume initially that no a_i is divisible by three. Define

$$a_i = 3A_i + s_i, \quad s_i \in \{1, 2\}, \quad Q = \sum_i A_i, \quad t = \#\{i : s_i = 2\}. \quad (6.1)$$

Then

$$S = 3Q + t. \quad (6.2)$$

Theorem 6.1 (Sharp spacing-three criterion). *The polynomial $P_{a,b,3}$ is unimodal if and only if*

$$(\exists i: 3 \mid a_i) \quad \text{or} \quad b \leq 1 + Q + 2 \left\lfloor \frac{t}{6} \right\rfloor, \quad (6.3)$$

where Q, t are as in (6.1) in the absence of a divisible factor. If no a_i is divisible by three and the inequality fails, write $t = 6h + v$, $0 \leq v \leq 5$, and $M = Q + 2h$. At the explicit index

$$n_0 = 3M + \left\lfloor \frac{v+3}{2} \right\rfloor \quad (6.4)$$

we have

$$n_0 \leq \lfloor D/2 \rfloor, \quad [q^{n_0}]P - [q^{n_0-1}]P = -1. \quad (6.5)$$

The proof has two independent parts. A binomial-prefix lemma proves sufficiency. A six-case residue table proves necessity, including the unit-drop certificate.

6.1 The binomial-prefix lemma

Lemma 6.2 (Enough ordinary binary factors). *For integers $B \geq 1$ and $t \geq 0$, the polynomial*

$$(1 + q)^t [B]_{q^3}$$

is unimodal whenever $t \geq 6\lfloor B/2 \rfloor$.

Proof. For $B = 1$ this is an ordinary binomial polynomial. Suppose $B \geq 2$, put $m = \lfloor B/2 \rfloor$ and $K = 6m$, and first take $t = K$. Thus $B = 2m$ or $B = 2m + 1$. Let

$$f_j = \binom{K}{j}, \quad u_n = \sum_{\ell \geq 0} f_{n-3\ell}, \quad d_n = u_n - u_{n-1},$$

with $f_j = 0$ outside $0 \leq j \leq K$ and $u_n = d_n = 0$ for $n < 0$. Let R_0, R_1, R_2 be the full residue masses of $(1 + q)^K$.

Modulo $\Phi_3(q) = 1 + q + q^2$, one has $(1 + q)^3 \equiv -1$ and hence $(1 + q)^6 \equiv 1$. Since $6 \mid K$,

$$R_0 + R_1q + R_2q^2 \equiv 1 \pmod{\Phi_3(q)}.$$

Two polynomials of degree at most two congruent modulo Φ_3 differ by a constant multiple of Φ_3 . It follows that $R_0 - R_1 = 1$ and $R_1 = R_2$. Since $R_0 + R_1 + R_2 = 2^K$,

$$R_0 = \frac{2^K + 2}{3}, \quad R_1 = R_2 = \frac{2^K - 1}{3}. \quad (6.6)$$

Consequently

$$(\delta_0, \delta_1, \delta_2) = (1, -1, 0). \quad (6.7)$$

Symmetry $f_j = f_{K-j}$ gives a useful reflection identity. The terms of residue n omitted from u_n have indices $n + 3, n + 6, \dots$. Reflecting these indices about $K/2$ gives

$$u_n = R_{n \bmod 3} - u_{K-n-3} \quad (n \geq -1).$$

Subtract the same identity at $n - 1$. For $n \geq 0$ this yields

$$d_n = \delta_{n \bmod 3} + d_{K-n-2}. \quad (6.8)$$

For $0 \leq n \leq K/2$,

$$d_n = \sum_{\ell \geq 0} (f_{n-3\ell} - f_{n-1-3\ell}) \geq 1.$$

Every summand is nonnegative because the binomial coefficients increase up to $K/2$: explicitly, $f_j/f_{j-1} = (K-j+1)/j > 1$ for $1 \leq j \leq K/2$. The summand with $\ell = 0$ is a positive integer. For $K/2 < n \leq K-2$, the reflected index $K-n-2$ lies between zero and $K/2$. Equations (6.7)–(6.8) therefore give $d_n \geq -1 + 1 = 0$. At the last two indices,

$$d_{K-1} = \delta_2 = 0, \quad d_K = \delta_0 = 1.$$

We have proved

$$d_n \geq 0 \quad (0 \leq n \leq K). \quad (6.9)$$

Now

$$(1 + q)^K [B]_{q^3} = (1 - q^{3B}) \sum_{n \geq 0} u_n q^n.$$

If $B = 2m$, its degree is $2K - 3$, its left-middle index is $K - 2$, and $3B = K$. If $B = 2m + 1$, its degree is $2K$, its middle index is K , and $3B = K + 3$. In either case the coefficients through the center agree with u_n , and their differences are nonnegative by (6.9). Symmetry proves unimodality for $t = K$. Finally, multiplication by $(1 + q)^{t-K}$ preserves SU (Lemma 2.2), proving the claim for every $t \geq K$. $\square$

6.2 Sufficiency of the corrected criterion

Proof of Theorem 6.1, sufficient direction. A divisible factor is handled by Lemma 2.3. Otherwise assume the right-hand inequality in (6.3). For $b \leq Q + 1$, Theorem 3.2 applies. For $b > Q + 1$, use Corollary 3.3 with $r = 3$. The residue product is

$$(1 + q)^t [b - Q]_{q^3}.$$

Set $B = b - Q$. The assumed inequality says $B \leq 1 + 2\lfloor t/6 \rfloor$, which implies $t \geq 6\lfloor B/2 \rfloor$. Lemma 6.2 proves that the residue product is SU, and Corollary 3.3 lifts this back to the original product. $\square$

6.3 Necessity and the coefficient-drop certificate

Proof of Theorem 6.1, necessary direction. Suppose no a_i is divisible by three, and write $t = 6h + v$, $0 \leq v \leq 5$, and $M = Q + 2h$. Then $S = 3M + v$ by (6.2).

Let R_0, R_1, R_2 be the residue masses of $F = \prod_i [a_i]_q$ modulo three. Complete blocks of three terms vanish modulo $\Phi_3(q) = 1 + q + q^2$, so

$$F(q) \equiv \prod_i [s_i]_q = (1 + q)^t \equiv (1 + q)^v \pmod{\Phi_3(q)}. \quad (6.10)$$

The degree-at-most-two residue polynomials of F and $(1 + q)^v$ are congruent modulo Φ_3 , so they differ by a constant multiple of Φ_3 . Their corresponding residue differences are therefore equal. The binomial theorem for $v = 0, \dots, 5$ gives Table 2.

v	δ_0	δ_1	δ_2	$e = \lfloor (v+3)/2 \rfloor$	$n_0 \bmod 3$	$n_0 - S$
0	1	-1	0	1	1	1
1	1	0	-1	2	2	1
2	0	1	-1	2	2	0
3	-1	1	0	3	0	0
4	-1	0	1	3	0	-1
5	0	-1	1	4	1	-1

Table 2: The residue class selected by $n_0 = 3M + e$ always has difference -1 .

If the bound in (6.3) fails, then $b \geq M + 2$. Choose n_0 from (6.4). The table shows $n_0 \geq S - 1$ and $n_0 \geq 1$. Moreover,

$$n_0 \leq 3M + 4 < 3M + 6 \leq 3b.$$

Thus Lemma 4.1 applies at n_0 , and the selected table entry gives the exact difference -1 . Finally,

$$\begin{aligned} \left\lfloor \frac{D}{2} \right\rfloor &= \left\lfloor \frac{3M + v + 3(b-1)}{2} \right\rfloor \\ &\geq \left\lfloor \frac{6M + v + 3}{2} \right\rfloor = 3M + \left\lfloor \frac{v+3}{2} \right\rfloor = n_0. \end{aligned}$$

The negative difference occurs in the first half of the symmetric sequence, contradicting (2.1). This proves both necessity and (6.5). $\square$

6.4 Consequences, minimality, and the missing sixth factor

Corollary 6.3 (A two-parameter infinite family). *For $t \geq 0$ and $B \geq 1$,*

$$(1 + q)^t [B]_{q^3} \text{ is unimodal} \iff t \geq 6 \left\lfloor \frac{B}{2} \right\rfloor. \quad (6.11)$$

Proof. For $t \geq 1$, take t ordinary factors $a_i = 2$ in Theorem 6.1, so $Q = 0$. The condition becomes $B \leq 1 + 2\lfloor t/6 \rfloor$, equivalent to (6.11). For $t = 0$, the spaced polynomial is unimodal only for $B = 1$; for $B \geq 2$ its first coefficients include 1, 0 and a later 1. $\square$

Every pair $B \geq 2, t \geq 6\lfloor B/2 \rfloor$ supplies a counterexample to the original necessity assertion: the source condition would require $B \leq 1$. Thus the degree-nine example is not an isolated coincidence, and the family is genuinely two-parameter. Corollary 7.8 below strengthens the one-parameter slice $B = 2$ to log-concavity; whether log-concavity also holds throughout the wider two-parameter family of (6.11) is not settled here and would be a natural next check. The two families are kept side by side; see item D4 of Table 6.

Corollary 6.4 (Minimality in the spacing-three branch). *Among counterexamples to the original necessity assertion with $r = 3$, at least six nontrivial ordinary factors are required, and the total polynomial degree is at least nine. Both bounds are attained by $(1 + q)^6(1 + q^3)$.*

Proof. A counterexample has no divisible factor and satisfies

$$Q + 1 < b \leq Q + 1 + 2\lfloor t/6 \rfloor.$$

Hence $t \geq 6$, so there are at least six factors of residue two, each nontrivial. Also $S = 3Q + t \geq 6$ and $b \geq 2$, whence $D = S + 3(b - 1) \geq 9$. The displayed example realizes equality. $\square$

This is a statement about the $r = 3$ branch only; it asserts nothing about other spacings. Proposition 8.1 closes exactly that gap by a different method, and the two results are retained together; see item D3 of Table 6.

For $k \leq 5$, one necessarily has $t \leq 5$, so the correction $2\lfloor t/6 \rfloor$ vanishes. The corrected theorem therefore agrees exactly with the old spacing-three criterion throughout that range. This explains the relevance of the source's reported cutoff of five ordinary factors [1, p. 14]; it is not a numerical accident.

Ordinary factors	Q	t	Old bound	Exact bound	Extra admissible b
$[2]_q^6$	0	6	1	3	2, 3
$[2]_q^{12}$	0	12	1	5	2, 3, 4, 5
$[5]_q^6$	6	6	7	9	8, 9
$[2]_q^5[4]_q$	1	5	2	2	None

Table 3: The exact bound means the largest b for which the product with $[b]_{q^3}$ is unimodal.

The mechanism is arithmetic: modulo Φ_3 , a residue-two factor is $1 + q$, whose sixth power is one. Each complete group of six such factors allows two additional terms in the spaced factor. In contrast, for $r = 2$ the only nonzero residue is one, so no analogous correction exists.

7 The two-shift family: an exact smoothing threshold

The smallest counterexample lives in the slice $b = 2, a_1 = \dots = a_n = 2$ of the general product (1.1). That slice is small enough to classify completely for *every* spacing, and the classification explains the special role of the exponent six at $r = 3$ from a second, entirely independent direction. Write

$$\mathcal{B}_{n,r}(q) = (1 + q)^n(1 + q^r) = \sum_{j=0}^{n+r} c_j^{(n,r)} q^j, \quad c_j^{(n,r)} = \binom{n}{j} + \binom{n}{j-r}, \quad (7.1)$$

with binomial coefficients taken to be zero when the lower index leaves $\{0, \dots, n\}$. In the notation of (1.1), $\mathcal{B}_{n,r} = P_{(2,\dots,2),2,r}$ with n ordinary factors.

7.1 A weighted subset model

An object is a pair (S, ε) with $S \subseteq \{1, \dots, n\}$ and $\varepsilon \in \{0, 1\}$, of weight

$$w(S, \varepsilon) = |S| + r\varepsilon.$$

Then $c_j^{(n,r)}$ counts objects of weight j . Complementation is a weight-reversing involution,

$$(S, \varepsilon) \mapsto (\{1, \dots, n\} \setminus S, 1 - \varepsilon), \quad w \mapsto n + r - w,$$

so that

$$c_j^{(n,r)} = c_{n+r-j}^{(n,r)}. \quad (7.2)$$

This is a weighted enumeration; it is not an assertion that w is the ordinary rank function on a product of two familiar graded posets.

7.2 The threshold

Theorem 7.1 (Sharp two-shift smoothing threshold). *For integers $r \geq 2$ and $n \geq 0$,*

$$\mathcal{B}_{n,r}(q) \text{ is unimodal} \iff n \geq r^2 - 3. \quad (7.3)$$

By Lemma 2.2 the set of exponents n for which $\mathcal{B}_{n,r}$ is unimodal is upward closed, so it suffices to prove failure at $n = r^2 - 4$ and unimodality at $n = r^2 - 3$, each by an argument valid for every r rather than by a numerical test. For $r = 2$ the two boundary polynomials are $1 + q^2$ and $1 + q + q^2 + q^3$, so the theorem holds; assume $r \geq 3$ from here on. Sections 7.3 and 7.4 supply the two halves, and Section 7.5 gives a second, independent proof of the harder half. The theorem is proved once; everything labelled “alternative” or “cross-check” below is a redundant confirmation, retained deliberately (items D1 and D2 of Table 6).

7.3 The lower bound: a central dip one step below the threshold

A symmetric unimodal sequence cannot have a central entry smaller than its neighbour. For $\mathcal{B}_{n,r}$ that obstruction factors, and the factorization is valid for every admissible n , not only at the boundary exponent.

Lemma 7.2 (Central difference in even total degree). *Suppose $n \geq r$ and $n + r = 2m$. Then*

$$c_m^{(n,r)} - c_{m-1}^{(n,r)} = \binom{n}{m} \frac{n + 2 - r^2}{(m + 1)(n - m + 1)}. \quad (7.4)$$

Proof. By binomial symmetry, $c_m = 2\binom{n}{m}$ and $c_{m-1} = \binom{n}{m-1} + \binom{n}{m+1}$. Divide by the positive quantity $\binom{n}{m}$ and use consecutive binomial ratios:

$$\frac{c_m - c_{m-1}}{\binom{n}{m}} = 2 - \frac{m}{n - m + 1} - \frac{n - m}{m + 1} = \frac{n + 2 - r^2}{(m + 1)(n - m + 1)},$$

the last step using $2m = n + r$. □

The single factor $n + 2 - r^2$ carries the whole sign, which is where the quadratic scale r^2 comes from algebraically. Now put $n_- = r^2 - 4$. For $r \geq 3$ we have $n_- \geq r$, and $n_- + r$ is even, so (7.4) applies and gives

$$c_m^{(n_-,r)} - c_{m-1}^{(n_-,r)} = -\frac{2\binom{r^2-4}{m}}{(m + 1)(r^2 - 3 - m)} < 0, \quad m = \frac{r^2 + r - 4}{2}. \quad (7.5)$$

Hence $\mathcal{B}_{r^2-4,r}$ is not unimodal. If $\mathcal{B}_{n,r}$ were unimodal for some $n < r^2 - 4$, repeated application of Lemma 2.2 would make $\mathcal{B}_{r^2-4,r}$ unimodal, a contradiction. This proves the whole necessary bound $n \geq r^2 - 3$, including the small values of n at which the two shifted binomial rows have gaps or disjoint supports.

That last deduction avoids a real edge-case error: applying (7.4) where $\binom{n}{m} = 0$ would not prove a strict central dip. The factorization is therefore used only where its positivity hypotheses hold, and the obstruction is propagated downward by contraposition of a valid smoothing lemma.

Remark 7.3 (Independent boundary cross-check, D2). The boundary case of (7.5) can also be obtained directly, without the general lemma. Put $N = n_- + 1 = r^2 - 3$ and $u = (N - r - 1)/2$. Writing the central difference through the first-difference function (7.7) below and reflecting, it equals $\Delta_u - \Delta_{u+1}$, and $\binom{N}{u+1} = \binom{N}{u}(N - u)/(u + 1)$ gives

$$\Delta_u - \Delta_{u+1} = \frac{\binom{N}{u}}{N} \left((r + 1) - (r - 1) \frac{N + r + 1}{N - r + 1} \right) = -\frac{4\binom{N}{u}}{N(N - r + 1)} < 0, \quad (7.6)$$

the denominator being positive for $r \geq 3$. One checks $u = (r^2 - r - 4)/2 = m - r$, and the two expressions agree: at $r = 3$ both equal -1 , at $r = 4$ both equal -22 , at $r = 5$ both equal -3230 , matching `data/thresholds.csv`. The general lemma is the more useful statement; the one-line boundary computation is an independent confirmation of the same sign.

7.4 The upper bound: a monotone ratio certificate

It remains to prove unimodality at $n = r^2 - 3$. The difficulty is that near the midpoint of the whole polynomial one binomial row is already decreasing while its translate is still increasing, so their first differences must be compared quantitatively.

Fix n , put $N = n + 1$, and define for every integer j

$$\Delta_j = \binom{n}{j} - \binom{n}{j-1} = \frac{N - 2j}{N} \binom{N}{j} \quad (0 \leq j \leq N). \quad (7.7)$$

Consequently

$$\Delta_j > 0 \quad (0 \leq j < \frac{N}{2}), \quad \Delta_{N-j} = -\Delta_j, \quad (7.8)$$

and, whenever both sides are positive, consecutive ratios are

$$h(j) := \frac{\Delta_{j+1}}{\Delta_j} = \frac{(n + 1 - j)(n - 1 - 2j)}{(j + 1)(n + 1 - 2j)}. \quad (7.9)$$

The coefficient differences of $\mathcal{B}_{n,r}$ satisfy

$$c_j^{(n,r)} - c_{j-1}^{(n,r)} = \Delta_j + \Delta_{j-r}. \quad (7.10)$$

The next identity is the main calculation. It is stated with a free n ; only its specialization to $n = r^2 - 3$ is used.

Lemma 7.4 (Ratio identity). *Let $y = n + 1 - r - \tau$. Where the indicated denominators are nonzero,*

$$h(\tau)h(y-1) - 1 = -\frac{r(n+1)(n+2-r^2+(n-r-2\tau)^2)}{(\tau+1)y(n+1-2\tau)(n+3-2y)}. \quad (7.11)$$

Proof. By (7.9) the left side equals

$$\frac{(n+1-\tau)(n-1-2\tau)(n+2-y)(n+1-2y)}{(\tau+1)y(n+1-2\tau)(n+3-2y)} - 1.$$

Place the two terms over the common denominator and substitute $y = n + 1 - r - \tau$. The resulting numerator factors as $-r(n+1)(n+2-r^2+(n-r-2\tau)^2)$, which is the claim. After clearing denominators this is a polynomial identity; no approximation is involved. $\square$

Specialization at the threshold. Set $n = r^2 - 3$ and

$$T = \frac{n+1-r}{2} = \frac{r^2-r-2}{2}, \quad (7.12)$$

an integer because $r(r-1)$ is even. The degree $n+r$ is odd and the left midpoint of $\mathcal{B}_{n,r}$ is

$$\mu = \left\lfloor \frac{n+r}{2} \right\rfloor = T + r - 1. \quad (7.13)$$

Let $1 \leq j \leq \mu$. If $j \leq (n+1)/2$, both terms of (7.10) are nonnegative, with $\Delta_{j-r} = 0$ when $j < r$; that part of the left half is already controlled. Suppose instead

$$\frac{n+1}{2} < j \leq \mu, \quad (7.14)$$

and write

$$\tau = j - r, \quad y = n + 1 - j = 2T - \tau.$$

Then $0 \leq \tau < T < y < (n+1)/2$; for $\tau \geq 0$ note that $(n+1)/2 = (r^2-2)/2 \geq r$ when $r \geq 3$. Hence Δ_τ and Δ_y are both positive and, by (7.8) and (7.10),

$$c_j - c_{j-1} = \Delta_\tau - \Delta_y. \quad (7.15)$$

Define, throughout this positive range and extended to $\tau = T$,

$$\rho_\tau = \frac{\Delta_\tau}{\Delta_{2T-\tau}}, \quad \text{so that} \quad \rho_T = 1.$$

For $\tau < T$,

$$\frac{\rho_{\tau+1}}{\rho_\tau} = \frac{\Delta_{\tau+1}}{\Delta_\tau} \frac{\Delta_{2T-\tau}}{\Delta_{2T-\tau-1}} = h(\tau)h(y-1). \quad (7.16)$$

All four factors of the denominator in (7.11) are positive here, and at $n = r^2 - 3$ its parenthesized numerator becomes

$$n + 2 - r^2 + (n - r - 2\tau)^2 = -1 + (2(T - \tau) - 1)^2 = 4(T - \tau)(T - \tau - 1) \geq 0 \quad (\tau \leq T - 1). \quad (7.17)$$

Therefore $\rho_{\tau+1} \leq \rho_\tau$, and iterating towards the endpoint gives $\rho_\tau \geq \rho_T = 1$. By (7.15) every remaining left-half difference is nonnegative, so $\mathcal{B}_{r^2-3,r}$ is unimodal. Together with Section 7.3 and Lemma 2.2, this proves Theorem 7.1. Note that the argument controls *every* left-half difference; positivity of a central difference alone would not have sufficed.

7.5 An alternative proof of the upper bound

The following is a second, self-contained proof of unimodality at $n = r^2 - 3$. It compares the same two first differences, but runs the induction in the opposite direction, from a different anchor, and produces a different factorization. Neither proof is a transcription of the other, and each is complete on its own; keeping both is deliberate (item D1 of Table 6).

Second proof of unimodality at $n = r^2 - 3$. Here $N = n + 1 = r^2 - 2$ and the degree $n + r$ is odd. For a first-half index $j \leq N/2$ both terms of (7.10) are nonnegative. For the remaining first-half indices put

$$s = N + r - 2j, \quad u = j - r = \frac{N - r - s}{2}, \quad v = N - j = \frac{N - r + s}{2}.$$

Because $j > N/2$ and $j \leq (n + r - 1)/2$, the integer s is even and satisfies $2 \leq s < r$. Furthermore $0 \leq u < v < N/2$, so $\Delta_u, \Delta_v > 0$, and by reflection (7.10) equals $\Delta_u - \Delta_v$. It therefore suffices to prove

$$\varrho_s := \frac{\Delta_u}{\Delta_v} = \frac{r+s}{r-s} \cdot \frac{\binom{N}{(N-r-s)/2}}{\binom{N}{(N-r+s)/2}} \geq 1. \quad (7.18)$$

At $s = 2$,

$$\varrho_2 = \frac{(r+2)(N-r)(N-r+2)}{(r-2)(N+r)(N+r+2)} = 1,$$

because the numerator minus the denominator is

$$4N(N - r^2 + 2) = 0. \quad (7.19)$$

For an even s with $s + 2 < r$, the adjacent ratio is

$$\frac{\varrho_{s+2}}{\varrho_s} = \frac{(r+s+2)(r-s)(N-r-s)(N-r+s+2)}{(r+s)(r-s-2)(N+r+s+2)(N+r-s)}. \quad (7.20)$$

All factors in the denominator are positive. Expanding the numerator minus the denominator gives

$$4Nr(N - r^2 + s^2 + 2s + 2) = 4Nr s(s + 2) > 0, \quad (7.21)$$

where the equality uses $N = r^2 - 2$. Starting from $\varrho_2 = 1$, induction over the even values of s proves (7.18). All first-half differences are nonnegative, and symmetry gives unimodality. $\square$

Remark 7.5 (How the two routes differ). Under $s = 2(T - \tau)$ the two comparisons are the same ratio, but the inductions are genuine mirror images with different content. The proof of Section 7.4 anchors at $\tau = T$, where $\rho_T = \Delta_T/\Delta_T = 1$ holds tautologically, and steps downward in s using the free- n identity (7.11) specialized by (7.17). The proof above anchors one step further in, at $s = 2$, where equality is not a tautology but the nontrivial identity (7.19), and steps upward in s using (7.21). Only the first route states its identity for arbitrary n , which is what makes the equality analysis of Section 7.6 fall out of it; only the second avoids introducing the auxiliary variable y . Their algebraic certificates, (7.19)–(7.21) versus (7.11)–(7.17), are checked by two different computer algebra systems in Section 10.3.

7.6 Equality, plateaus, and log-concavity

The equality case is visible in the first proof, with no extra asymptotics or coefficient calculations.

Corollary 7.6 (Exactly four maxima at the threshold). *Let $r \geq 2$, $n = r^2 - 3$ and $\mu = (n + r - 1)/2$. The coefficients of $\mathcal{B}_{n,r}$ attain their maximum exactly at*

$$\mu - 1, \mu, \mu + 1, \mu + 2, \quad (7.22)$$

and are strictly increasing before this block and strictly decreasing after it. Empty flanks, as occur for $r = 2$, are allowed.

Proof. For $r = 2$ the polynomial is $1 + q + q^2 + q^3$. Assume $r \geq 3$. In (7.17) equality holds at $\tau = T - 1$ and the factor is strictly positive at every earlier integer of the comparison range. Hence $\rho_{T-1} = \rho_T = 1$ while $\rho_\tau > 1$ for $\tau < T - 1$, so the only vanishing left-half difference is at $j = (T - 1) + r = \mu$. In the easy range $j \leq (n + 1)/2$ the first term Δ_j is strictly positive unless $j = (n + 1)/2$ is an integer; at that exceptional index $\Delta_{j-r} > 0$, so the sum is still strictly positive. Consequently $c_{\mu-1} = c_\mu$ and all preceding differences are positive. Odd-degree symmetry gives $c_\mu = c_{\mu+1}$ and $c_{\mu+2} = c_{\mu-1}$. $\square$

The plateau then contracts under further smoothing. By (2.3), a flat maximum block of length $w \geq 2$ with strict flanks becomes a block of length $w - 1$, and a unique maximum becomes a two-term central block. Starting at $n = r^2 - 3$, the numbers of maximizing coefficients are therefore

$$4, 3, 2, 1, 2, 1, 2, \dots \quad (7.23)$$

as n increases by one; the endpoints of each block are fixed by symmetry and the total degree.

For the next statement, call a nonnegative finite sequence *log-concave* when it has no internal zeros and $a_j^2 \geq a_{j-1}a_{j+1}$ at every internal index. The support condition matters: local inequalities alone can hold vacuously across a sufficiently long gap.

Lemma 7.7 (($1 + q$) preserves log-concavity). *Multiplication by $1 + q$ preserves log-concavity of a positive coefficient sequence.*

Proof. Extend a_j by zero outside its support and set $b_j = a_j + a_{j-1}$. Direct expansion gives

$$b_j^2 - b_{j-1}b_{j+1} = (a_j^2 - a_{j-1}a_{j+1}) + (a_{j-1}^2 - a_{j-2}a_j) + (a_{j-1}a_j - a_{j-2}a_{j+1}). \quad (7.24)$$

The first two summands are nonnegative by log-concavity. In the interior, log-concavity says that the positive consecutive ratios a_j/a_{j-1} are nonincreasing; in particular $a_{j-1}/a_{j-2} \geq a_{j+1}/a_j$, which makes the third summand nonnegative. At the endpoints the same claim follows directly from the zero extension, and the new support has no gaps. $\square$

This is a special case of familiar convolution closure phenomena; see [3] for broader context. The elementary identity above supplies all that is needed here.

Corollary 7.8 (A log-concave infinite family). *For every $n \geq 6$, the polynomial $[2]_q^n [2]_{q^3} = (1 + q)^n (1 + q^3)$ is a log-concave counterexample to the necessity assertion of [1, Conjecture 5.4].*

Proof. At $n = 6$ the eight interior quantities $c_j^2 - c_{j-1}c_{j+1}$ are

$$21, 99, 126, 0, 0, 126, 99, 21, \quad (7.25)$$

so the witness is log-concave, and Lemma 7.7 propagates this to every larger n . Theorem 7.1 supplies unimodality. For every member $r = 3$, no $a_i = 2$ is divisible by r , and the floor sum in (1.3) is zero, so the required inequality would read $2 \leq 1$. $\square$

Corollaries 6.3 and 7.8 are kept together: the first is broader in parameters, the second stronger for each member (item D4 of Table 6). For $r = 3$, Theorem 7.1 gives the threshold $n = 6$, in agreement with Corollary 6.3 at $B = 2$: two independent routes to the same special value six.

7.7 Threshold values and a scope warning

The first thresholds are consequences of Theorem 7.1, not extrapolations from the table.

Gap r	First unimodal exponent $r^2 - 3$	Degree	Four mode indices
2	1	3	0–3
3	6	9	3–6
4	13	17	7–10
5	22	27	12–15
6	33	39	18–21
7	46	53	25–28
8	61	69	33–36
9	78	87	42–45
10	97	107	52–55

For $r \geq 4$ these members still violate the floor-sum condition (1.3), but they have both $k = n > 3$ and $r > 3$. They are therefore *not* additional counterexamples inside the conjecture's necessity range; they describe the smoothing phenomenon beyond that range.

8 Minimum degree across all spacings

A small counterexample need not be the smallest. Corollary 6.4 proved, by theory and without computation, that within the $r = 3$ branch at least six nontrivial ordinary factors and degree at least nine are required, and explicitly said nothing about other spacings. This section closes that gap by a complete finite search over *all* spacings, without imposing arbitrary bounds on k or on the factor lengths.

8.1 Reduction to finitely many normalized parameter sets

For a counterexample to necessity we may assume $b \geq 2$, since $b = 1$ always satisfies (1.3). Factors $[1]_q = 1$ can be deleted: deleting them changes neither the polynomial nor C_r , and cannot invalidate the applicability of the necessity assertion, because $k \leq 3$ remains true when k decreases and $r \leq 3$ is untouched. At least one nontrivial ordinary factor must remain, since otherwise the polynomial is $[b]_{q^r}$, which is not unimodal for $r, b \geq 2$. We may therefore normalize by

$$2 \leq a_1 \leq a_2 \leq \cdots \leq a_k.$$

For fixed D , every normalized candidate then satisfies

$$2 \leq r \leq D - 1, \quad 2 \leq b \leq 1 + \left\lfloor \frac{D - 1}{r} \right\rfloor, \quad (8.1)$$

and once r, b are fixed the positive integers $a_i - 1$ form a partition of $S = D - r(b - 1) \geq 1$. Conversely each such partition gives exactly one normalized parameter set. Hence the enumeration is complete and contains no permutation duplicates.

8.2 The certificate

Proposition 8.1 (Exact finite computation). *Among products (1.1) in the asserted necessity range, the minimum degree of a counterexample to necessity is nine. After deleting trivial factors and permuting the ordinary factors, the degree-nine counterexample parameter set is unique, namely $r = 3$, $b = 2$, $a_1 = \cdots = a_6 = 2$.*

Finite verification and its completeness argument. The reductions above exhaust all possible counterexamples of smaller degree. Running two independent coefficient implementations over those parameter sets gives the counts below.

Degree	Normalized parameter sets	Necessity counterexamples	Sufficiency counterexamples
1	0	0	0
2	0	0	0
3	1	0	0
4	3	0	0
5	7	0	0
6	13	0	0
7	23	0	0
8	38	0	0
9	59	1	0
Total	144	1	0

The single flagged row has $r = 3$, $b = 2$ and six factors of length two. The complete rows, including all coefficient arrays and both logical flags, are in `data/degree_search.csv`; the executable enumeration is `code/verify_threshold.py`. Appendix B gives a second, compact, standalone realization of the same search. $\square$

The zero column for sufficiency is consistent with Theorem 3.2, which proves sufficiency outright; it is a bounded computational observation, not the reason sufficiency holds. This is a computer-checked finite enumeration, not a proof-assistant formalization: its finite coverage is proved above, its arithmetic is integer-exact, and its whole state space is small enough to inspect. The threshold theorem of Section 7 is independent of it.

Proposition 8.1 and Corollary 6.4 are kept side by side because neither implies the other (item D3 of Table 6). The corollary explains *why* six residue-two factors are forced at spacing three, by an argument valid without any computation; the proposition establishes that no other spacing does better, which no part of the $r = 3$ analysis can see.

9 Why a quadratic threshold is natural: a heuristic

This section is explanatory only. Nothing in it is used in any proof above, and none of it is offered as evidence for Theorem 7.1.

Choose a weighted subset object of Section 7 uniformly at random. Its weight is $X + rB$ with $X \sim \text{Bin}(n, 1/2)$ and $B \sim \text{Bernoulli}(1/2)$ independent, and its probability generating function is $2^{-(n+1)}\mathcal{B}_{n,r}(q)$. Conditionally on B it is one of two translated binomial distributions, with means separated by r and with component standard deviation $\sqrt{n}/2$.

There is an instructive continuous analogue. Up to a positive normalization, a symmetric mixture of two normal densities with variance σ^2 and means $\pm a$ has density

$$f(x) = e^{-(x-a)^2/(2\sigma^2)} + e^{-(x+a)^2/(2\sigma^2)} = 2e^{-(x^2+a^2)/(2\sigma^2)} \cosh\left(\frac{ax}{\sigma^2}\right),$$

so that for $x > 0$

$$\frac{f'(x)}{f(x)} = \frac{-x + a \tanh(ax/\sigma^2)}{\sigma^2}.$$

If $a \leq \sigma$, then $\tanh u \leq u$ for $u \geq 0$ gives $a \tanh(ax/\sigma^2) \leq a^2x/\sigma^2 \leq x$, so the density is nonincreasing on the positive half-line and is unimodal. If $a > \sigma$, differentiation gives $f''(0)/f(0) = (a^2 - \sigma^2)/\sigma^4 > 0$, so the centre is a local minimum rather than a mode. The continuous threshold is therefore exactly $a = \sigma$. Putting $a = r/2$ and $\sigma = \sqrt{n}/2$ predicts the scale $n \approx r^2$, against the exact discrete answer $n \geq r^2 - 3$.

The quadratic scale thus has a simple variance interpretation, while the integer correction and the four-term equality plateau require the exact calculation. Approximation by normal densities cannot rule out small local coefficient dips, and weak convergence of distributions does not preserve unimodality in the direction a finite coefficient theorem needs.

10 Exact computations and reproducibility

The proofs above establish the infinite statements. Two separate exact verifiers check finite cases using Python integers, without floating-point arithmetic or external packages. `code/verify_products.py` covers the general product (1.1); the run recorded in `data/verification_results.json` passed every check of Table 4. `code/verify_threshold.py` covers the two-shift family and the minimum-degree search; the run recorded in `data/verification_report.json` passed every check of Table 5. The two suites have disjoint test families and disjoint programming interfaces, and neither imports the other.

Check in code/verify_products.py	Number of cases
Independent schoolbook versus optimized multiplication	1,000
Centered peeling identity on an integer grid	3,003
Exhaustive spacing-three multiset products	30,539
Deterministic random spacing-three products, no divisible factor	30,000
Spacing-three products with a divisible factor	2,000
General-spacing sufficient condition	10,000
General-spacing necessary degree bound	10,000
Spacing-two necessary-and-sufficient criterion	5,000
Two-shift boundary polynomials, $2 \leq r \leq 60$	177
Exact two-shift negative central-difference formula	58
Spacing-three binomial-comb threshold samples	237

Table 4: Finite checks actually executed by the general-product verifier. Some families overlap; the counts are not numbers of distinct mathematical theorems or independent proofs.

Check in code/verify_threshold.py	Count	Range or purpose
Full rectangular threshold grid	3,263	$2 \leq r \leq 14, 0 \leq n \leq 250$
Boundary and nearby rows	594	$2 \leq r \leq 100, n = r^2 - 3 + \delta, \delta = -1, \dots, 4$
Central-difference identity	98	One exact obstruction for each $3 \leq r \leq 100$
Ratio comparisons	2,450	All comparisons used at the tested thresholds
Minimum-degree parameter sets	144	Every normalized case with $D \leq 9$
Independent tuple enumerations	144	4,762 objects in total

Table 5: Finite checks actually executed by the two-shift verifier. Some boundary cases also occur in the rectangular grid; these are test counts, not a claim that every test involves a distinct parameter pair.

10.1 What was checked

The exhaustive spacing-three test considers all multisets of ordinary parameters from $\{1, 2, 4, 5, 7, 8\}$, with zero through eight factors, and all b from one through the corrected bound plus two. The zero-factor cases check the harmless empty-product boundary. In this finite collection, 938 products violate the old necessity condition while satisfying the corrected theorem. Every predicted nonunimodal case is also checked at the explicit index (6.4); the drop is exactly -1 , not merely negative.

The separate random spacing-three experiment uses seed 1977, chooses k uniformly from $1, \dots, 16$, chooses each a_i from the integers $1, \dots, 69$ not divisible by three, and chooses b uniformly from one through the corrected bound plus eight. The seed is reset for this experiment so it is independent of how many preliminary tests are run.

The two-shift boundary checks additionally verify the plateau lengths 4, 3, 2, 1, 2 predicted by (7.23) and the strictness of the threshold's left flank.

10.2 Five coefficient routes and two unimodality tests

The small counterexample is expanded by three mutually independent routes in `code/verify_products.py` alone: the sliding-sum recurrence, the binomial theorem, and direct enumeration of all $2^7 = 128$ choices of six weights one and one weight three. Across the package, five distinct coefficient implementations are retained (item D7 of Table 6):

- (i) the sliding-sum recurrence

$$v_j = u_j + v_{j-s} - u_{j-as}, \quad (10.1)$$

obtained by extracting coefficients from $(1 - q^s)B(q) = (1 - q^{as})A(q)$, where $B = [a]_{q^s} A$;

- (ii) schoolbook convolution, cross-checked against the optimized version in 1,000 cases;
- (iii) direct tuple enumeration over $(x_1, \dots, x_k, z)$ with $0 \leq x_i < a_i$ and $0 \leq z < b$, incrementing the coefficient at $x_1 + \dots + x_k + rz$, using no polynomial multiplication at all (144 cases, 4,762 tuples);
- (iv) the binomial theorem for the two-shift family; and
- (v) the 128-element binary-choice enumeration of the specific counterexample.

Unimodality itself is tested in two ways throughout, and both are kept (item D8): a generic detector that looks for a rise after a descent and does *not* assume symmetry, and a test that uses symmetry plus the left-half first differences (2.1).

For a polynomial of degree D , one application of (10.1) costs $O(D)$ integer additions and subtractions, so a product with k ordinary factors can be formed in $O(kD)$ arithmetic operations and $O(D)$ working storage. These are arithmetic-operation bounds; the bit cost also depends on coefficient size. For the two-shift family a binomial row is generated by $\binom{n}{j+1} = \binom{n}{j}(n-j)/(j+1)$ with a checked exact division at each step, after which one shifted addition produces (7.1).

10.3 Two computer algebra witnesses

The two proofs of Theorem 7.1 rest on different algebraic certificates, and each was checked on a different system (item D6).

A Wolfram Language evaluation, using the connected Wolfram kernel reporting version 15.0.1, confirmed the counterexample coefficients, the peeling identity (3.1), the six-case residue table of Table 2, the base identity (7.19) in the factored form $4nn(2 + nn - rr^2)$, the step identity (7.21) in the factored form $4nnrr(2 + nn - rr^2 + 2ss + ss^2)$, and the two-shift boundary

tests for $2 \leq r \leq 16$. The exact input and returned text are `code/verify_products.wl` and `data/wolfram_output.txt`.

A separate optional script, `code/symbolic_checks.py`, uses SymPy 1.14.0 to simplify three expressions to zero: the ratio identity (7.11), the central-difference identity (7.4), and the boundary factorization (7.17). Its output is `data/symbolic_report.json`; the main verifiers do not require SymPy.

Both are independent computational checks, not proof-assistant verification of the arguments, and the symbolic check is an audit of algebra rather than a substitute for establishing the positive-denominator hypotheses used in the proofs.

10.4 Two threshold tables on orthogonal axes

The package carries two threshold tables, neither a subset of the other (item D9). `data/r3_binomial_thresholds.csv` is indexed by $b = 1, \dots, 40$ and gives, for spacing three, the least unimodal exponent t of Corollary 6.3 together with the gap index and the gap value -1 at the preceding exponent. `data/thresholds.csv` is indexed by $r = 2, \dots, 100$ and gives `critical_n`, `below_n`, the exact `below_central_difference`, `critical_degree` and `plateau_start/plateau_end`. Both are regenerated by their respective verifiers.

10.5 Reproduction commands and output

From this directory, with Python 3.10 or later:

```
python3 code/verify_products.py --full --out data/verification_results.json
python3 code/verify_threshold.py
python3 code/minimal_certificate.py
python3 code/symbolic_checks.py # optional: requires SymPy
```

The first command writes a JSON run summary, the small counterexample certificate, and the spacing-three threshold CSV; it returns a nonzero exit code if any check fails. For a shorter smoke test, omit `-full` and choose a different output filename so that the recorded full run is preserved. The second command regenerates the JSON and CSV files of the two-shift analysis in `data/`. All checks use explicit exceptions rather than `assert`, so they remain active under `python -O`. No network access is used by any verification program.

Build this report with the supplied Makefile or run

```
latexmk -pdf -interaction=nonstopmode report.tex
```

A TeX installation with the packages listed in the preamble is needed, and `code/minimal_certificate.py` must be present because Appendix B typesets it directly. No custom font files, proprietary packages, BibTeX run or shell escape are required.

10.6 Closed-form decision versus coefficient expansion

Theorem 6.1 gives a decision procedure requiring $O(k)$ integer-arithmetic operations: test divisibility, compute Q and t , and compare b with the bound. If the answer is negative, (6.4) gives the witness index directly. No expansion of the polynomial is needed to decide the question. The actual bit complexity depends on the sizes of the input integers.

The positive proof uses induction and the centered identity. Literally unrolling all Q peeling steps can be long when the parameters are encoded in binary; no claim of a uniformly short expanded certificate is being made. The full finite verifier expands coefficients because it is intended as an independent check, not as the fastest implementation of the classification theorem.

10.7 Deliberate duplications

The merge that produced this note kept a number of results, proofs and artefacts in duplicate on purpose. Table 6 lists them, with one line each on what the redundancy buys.

Table 6: Results, proofs and artefacts deliberately kept in more than one copy.

	Duplicated item	What the second copy buys
D1	Two proofs of Theorem 7.1: Section 7.4 (free- n ratio identity, induction on τ down from $\rho_T = 1$) and Section 7.5 (induction on even s up from $\varrho_2 = 1$).	The first states its identity for arbitrary n and therefore yields the equality analysis for free; the second needs no auxiliary variable and anchors on a nontrivial identity, so a slip in either factorization cannot propagate to both.
D2	Two accounts of the failure at $n = r^2 - 4$: general Lemma 7.2 and the boundary computation (7.6).	The lemma exhibits $n + 2 - r^2$ as the sign-determining factor for every admissible n ; the one-line boundary computation confirms the same sign independently and agrees numerically.
D3	Two minimality results: Corollary 6.4 (theory, $r = 3$ only) and Proposition 8.1 (exhaustive, all r).	The first explains why six residue-two factors are forced and needs no computation; the second rules out every other spacing, which the $r = 3$ analysis cannot see. Neither implies the other.
D4	Two counterexample families: Corollary 6.3 $((1 + q)^t[B]_{q^3}$, two parameters) and Corollary 7.8 $((1 + q)^n(1 + q^3)$, log-concave).	The first is broader in parameters, the second stronger per member.
D5	Two witness polynomials: the plateau witness $(1 + q)^6(1 + q^3)$ and the strict witness $(1 + q)^9(1 + q^3)$.	The second has a unique maximum and strictly monotone flanks, answering the objection that the refutation is a plateau accident.
D6	Two computer algebra witnesses: Wolfram Language 15.0.1 (code/verify_products.wl) and SymPy 1.14.0 (code/symbolic_checks.py).	They check the two <i>different</i> algebraic routes of D1 on two different systems.
D7	Five coefficient implementations, listed in Section 10.	Agreement between a recurrence, a convolution, a pure counting enumeration and a closed form makes a shared implementation bug very unlikely.
D8	Two unimodality tests: generic rise-after-descent, and symmetry plus left-half differences.	The generic test does not assume symmetry, so it would catch an error in the symmetry reasoning itself.
D9	Two threshold tables: data/r3_binomial_thresholds.csv (axis b) and data/thresholds.csv (axis r).	Orthogonal axes; neither table's rows are derivable from the other's.
D10	Two closure lemmas: Lemma 2.1 (centered intervals) and Lemma 2.2 ($b_j - b_{j-1} = a_j - a_{j-2}$).	The first is the general tool the peeling induction needs, with its common-centre convention; the second is a short special case sufficient for Section 7, and it also supplies (7.23).
D11	Two provenance records, merged in PROVENANCE.md.	The union of search queries, pinpoints, dates and scope caveats from both original packages; see Section 11.

11 Provenance of this merged note

This note is the merge of two independently produced research packages that attacked the same conjecture and arrived at the same theorem.

docs/reports/log-concavity-and-unimodality/q-integer-product-unimodality (18 September 2026) treated the general product (1.1). It proved the shared threshold theorem as a supplementary result, through the first-difference function $H_n(j) = \binom{n}{j} - \binom{n}{j-1} = ((N-2j)/N)\binom{N}{j}$ with $N = n+1$, an induction over the even parameter $s = N+r-2j$ in $[2, r]$ on the ratio $H_n(u)/H_n(v)$, anchored at $s=2$ by the identity $4N(N-r^2+2)=0$ and stepped by $4Nr s(s+2) > 0$; and it computed the failure at $n=r^2-4$ as the single central difference $-4\binom{N}{u}/(N(N-r+1))$. That route survives here as Section 7.5 and Remark 7.3. The present note keeps that package's spine: Sections 2–6 and Sections 10–12 descend from it, including the centered peeling identity, the residue-mass obstruction, the spacing-two and spacing-three classifications, and the open-problem programme for $r \geq 4$.

docs/reports/log-concavity-and-unimodality/binomial-smoothing-threshold (20 September 2026) treated only the two-shift family, and made the shared threshold its main theorem. It proved it through $d_j = \binom{n}{j} - \binom{n}{j-1} = ((n+1-2j)/(n+1))\binom{n+1}{j}$, the consecutive ratio $h(j) = d_{j+1}/d_j$, and the free- n ratio identity (7.11), running an induction on $\tau \leq T = (r^2-r-2)/2$ over $R_\tau = d_\tau/d_{2T-\tau}$ anchored at $R_T = 1$ and using the specialization $4(T-\tau)(T-\tau-1) \geq 0$; and it proved the general central-difference Lemma 7.2 for every $n \geq r$ with $n+r$ even, specializing it at the boundary. That route is the primary proof here, Sections 7.3–7.4, because its free- n statement yields Corollary 7.6 directly. Sections 7, 8 and 9, and Appendices B and C, descend from that package.

Both packages refuted the source's necessity assertion with the identical witness and identical parameters, and both reduced the threshold to the two adjacent exponents $n = r^2-3$ and $n = r^2-4$ by the same upward-closure argument. The shared theorem is stated and proved once here, as Theorem 7.1; the second route is retained explicitly as an alternative proof. Search records, pinpoints and scope caveats from both packages are merged in PROVENANCE.md (item D11 of Table 6). Neither package was refereed or machine-verified, and no exhaustive priority determination was made by either.

12 Scope and further targets

The exact counterexample disposes of the stated necessity assertion. The stronger results retain and prove its sufficient half, classify all spacing-two and spacing-three products, classify the two-shift family for every spacing, and identify a precise residue obstruction rather than a merely numerical discrepancy. The proofs explain both the first counterexample and the complete families that it begins.

The source's main q -Fibonomial question is different. This note neither proves nor disproves that main conjecture, and it does not challenge the source's established small-case theorems. The counterexample addresses the auxiliary product criterion (1.3), not a claim that a particular q -Fibonomial coefficient is nonunimodal.

For $r \geq 4$, a general classification of $(\prod_i [a_i]_q)[b]_{q^r}$ is not obtained here. In particular, the necessity question for two or three ordinary factors at those spacings is not settled by the present arguments. Restricting the source's necessity clause to the surviving regimes would produce a new statement needing its own proof, not an automatic repair. What can be said precisely is this: necessity is *true* for $r=2$ (Theorem 5.1) and for $r=3$ whenever $k \leq 5$ (Theorem 6.1, since then $t \leq 5$), *false* for $r=3$ with $t \geq 6$, and *open* here for $k \leq 3$ with $r \geq 4$.

A concrete continuation is to combine the centered reduction with the full vector of residue differences at larger r . Lemma 4.1 gives exact obstruction indices whenever the periodic tail

enters the first half. Corollary 3.3 gives positive constructions from residue products. For $r = 3$, these two mechanisms meet exactly because $(1 + q)^6 \equiv 1 \pmod{\Phi_3(q)}$. For larger r , the residual factors $[s_i]_q$ range over more than 1 and $1 + q$, so the same one-parameter six-case reduction is no longer available. Determining when the positive and negative tests meet is a specific next task, not a conclusion asserted in this note.

Two further questions are left open by the merge itself: whether the wider two-parameter family of Corollary 6.3 is log-concave throughout, as the slice $B = 2$ is by Corollary 7.8; and whether the minimum-degree uniqueness of Proposition 8.1 persists at degrees ten and above, where the finite search was not run.

The structural lesson is narrower and more useful than a blanket rejection of arithmetic criteria: an invariant that assigns zero contribution to all short factors can fail to detect their cumulative smoothing effect. For Bernoulli factors that effect is measurable exactly — a gap of size r disappears at exponent $r^2 - 3$ — and the factored ratio identity explains both the threshold and its unusually flat equality case.

A A short standalone certificate

The disproof itself does not depend on any of the stronger theorems. The following elementary Python fragment checks it using only the binomial theorem.

```
from math import comb

c = [
    (comb(6, j) if 0 <= j <= 6 else 0)
    + (comb(6, j-3) if 0 <= j-3 <= 6 else 0)
    for j in range(10)
]
assert c == [1, 6, 15, 21, 21, 21, 21, 15, 6, 1]
assert c == c[::-1]
assert all(c[j-1] <= c[j] for j in range(1, 5))
a, b, r = [2]*6, 2, 3
assert not any(x % r == 0 for x in a)
assert b > 1 + sum(x/r for x in a)
assert len(a) <= 3 or r <= 3
```

The distributed full verifiers use explicit exceptions instead of `assert`, so running Python with optimization does not disable their checks. The fragment above is only a compact readable certificate.

B A compact independent minimum-degree verifier

The following program uses direct tuple enumeration only. It checks all 144 normalized cases through degree nine, prints their degree counts, and requires exactly the one stated violation. It has no third-party dependencies, and its explicit exception checks also remain active under Python's optimization flag. It is independent of `code/verify_threshold.py`: it shares no code with it and uses no polynomial multiplication.

```
#!/usr/bin/env python3
"""Standalone exhaustive degree <= 9 certificate; Python standard library."""
from itertools import product

def partitions(s, low=1):
    if s == 0:
        yield ()
```

```

    for u in range(low, s + 1):
        for tail in partitions(s - u, u):
            yield (u,) + tail

def unimodal(c):
    falling = False
    for x, y in zip(c, c[1:]):
        if y < x:
            falling = True
        if falling and y > x:
            return False
    return True

counts, failures = [], []
for degree in range(1, 10):
    count = 0
    for r in range(2, degree):
        for b in range(2, 2 + (degree - 1) // r):
            s = degree - r * (b - 1)
            for p in partitions(s):
                a = tuple(u + 1 for u in p)
                c = [0] * (degree + 1)
                for v in product(*(range(u) for u in a), range(b)):
                    c[sum(v[:-1]) + r * v[-1]] += 1
                condition = (any(u % r == 0 for u in a)
                             or b <= 1 + sum(u // r for u in a))
                applies = len(a) <= 3 or r <= 3
                if applies and unimodal(c) and not condition:
                    failures.append((degree, r, b, a))
            count += 1
    counts.append(count)
expected = [(9, 3, 2, (2, 2, 2, 2, 2, 2))]
if counts != [0, 0, 1, 3, 7, 13, 23, 38, 59]:
    raise RuntimeError("case counts", counts)
if failures != expected:
    raise RuntimeError("counterexamples", failures)
print("PASS:", sum(counts), "normalized cases")
print("Counts by degree:", counts)
print("Counterexamples:", failures)

```

C Recorded random area selection

The second of the two merged packages began from a recorded random draw over research areas. The initial tool execution was

```

index = secrets.randbelow(len(areas))
selected_area = areas[index]

```

It returned the zero-based index 0, selecting enumerative combinatorics. There was no redraw. The draw used Python's `secrets` module rather than a manually chosen seed. The raw operating-system entropy was not recorded; the realized index and the exact ordered list are saved in `data/area_selection.json`. Re-running the draw is not expected to reproduce the same index. All mathematical calculations in this note are deterministic.

Index	Area	Index	Area
0	Enumerative combinatorics	12	Algebraic combinatorics
1	Graph theory	13	Automata and formal languages
2	Combinatorics on words	14	Probability on finite structures
3	Finite geometry	15	Topological graph theory
4	Number theory	16	Extremal set theory
5	Permutation patterns	17	Polyhedral combinatorics
6	Order theory	18	Combinatorial game theory
7	Semigroup theory	19	Polynomial inequalities
8	Discrete geometry	20	Matrix theory
9	Additive combinatorics	21	Coding theory
10	Symbolic dynamics	22	Partition theory
11	Recreational integer sequences	23	Asymptotic analysis

References

- [1] Brendan B. Connelly, Ezekiel Ito, Thomas C. Martinez, Olha Shevchenko, and Kacey Yang. *Unimodality of q -Fibonomial coefficients for small cases*. arXiv:2605.12822v1 [math.CO], submitted 12 May 2026. [Version-pinned arXiv record](#); [HTML](#); [PDF](#). The target is Section 5.2, Conjecture 5.4 and the adjacent testing statement, p. 14.
- [2] Nantel Bergeron, Cesar Ceballos, and Josef Küstner. Elliptic and q -analogs of the Fibonomial numbers. *SIGMA* **16** (2020), paper 076, 16 pp. [doi:10.3842/SIGMA.2020.076](https://doi.org/10.3842/SIGMA.2020.076). Background for the distinction from the original Fibonomial question.
- [3] Richard P. Stanley. Log-concave and unimodal sequences in algebra, combinatorics, and geometry. *Annals of the New York Academy of Sciences* **576** (1989), 500–535. Author’s publication list, entry 72: math.mit.edu/~rstan/pubs.